

Question	Answer	Mark	Mark scheme	Additional guidance
16	(7, 3.5)	P1	process to find the gradient of OB , eg $\frac{-4-0}{2-0} (= -2)$	
		P1	process to find the equation of DE , eg substitutes $(12, -6.5)$ into $y = "-2"x + c$ or $y = [-2]x + c$ or $y - -6.5 = "-2"(x - 12)$ or $y - -6.5 = [-2](x - 12)$	Equation of DE is $y = -2x + 17.5$ Where $[-2]$ can be $\frac{-1}{2}$ or 2 or $\frac{1}{2}$ and must be labelled as the gradient of OB
		P1	process to find the equation of OD , eg substitutes $(0, 0)$ into $y = \frac{-1}{"-2"}x + c$ or $y = \frac{-1}{[-2]}x + c$ or $y - 0 = \frac{-1}{"-2"}(x - 0)$ or $y - 0 = \frac{-1}{[-2]}(x - 0)$	Equation of OD is $y = \frac{1}{2}x$ Where $[-2]$ can be $\frac{-1}{2}$ or 2 or $\frac{1}{2}$ and must be labelled as the gradient of OB
		P1	(dep P3) forms an equation that can be solved to find x coordinate of D , eg $"-2x + 17.5" = "\frac{1}{2}x"$	
		A1	cao	A correct answer with no supportive working gets 0 marks